

19) Create a Product Shopping Cart and Billing System

HTML product form, CSS table or card layout, JavaScript objects, arrays, functions, quantity processing, discount, tax and total calculation.

Code

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Shopping Cart and Billing System</title>


<style>

  * {

    box-sizing: border-box;

    margin: 0;

    padding: 0;

  }


  body {

    font-family: Arial, sans-serif;

    background: linear-gradient(135deg, #667eea, #764ba2);

    min-height: 100vh;

    padding: 30px;

  }


  .container {
```

```
    max-width: 900px;
    margin: auto;
    background: white;
    padding: 30px;
    border-radius: 15px;
    box-shadow: 0 10px 30px rgba(0,0,0,0.25);
}
```

```
h1 {
    text-align: center;
    color: #4b3f72;
    margin-bottom: 25px;
}
```

```
.product-form {
    display: grid;
    grid-template-columns: 2fr 1fr 1fr auto;
    gap: 10px;
    margin-bottom: 25px;
}
```

```
input {
    padding: 12px;
    border: 2px solid #ddd;
    border-radius: 8px;
    font-size: 15px;
}
```

```
button {  
    padding: 12px 18px;  
    border: none;  
    border-radius: 8px;  
    background: #667eea;  
    color: white;  
    cursor: pointer;  
    font-size: 15px;  
}
```

```
button:hover {  
    background: #4f63c4;  
}
```

```
table {  
    width: 100%;  
    border-collapse: collapse;  
    margin-top: 20px;  
}
```

```
th, td {  
    padding: 12px;  
    border: 1px solid #ddd;  
    text-align: center;  
}
```

```
th {  
    background: #4b3f72;
```

```
    color: white;
}
```

```
tr:nth-child(even) {
    background: #f5f5f5;
}
```

```
.quantity {
    width: 70px;
    padding: 7px;
    text-align: center;
}
```

```
.remove {
    background: #e63946;
    padding: 7px 12px;
}
```

```
.billing {
    margin-top: 25px;
    margin-left: auto;
    max-width: 350px;
    background: #f4f4f4;
    padding: 20px;
    border-radius: 10px;
}
```

```
.billing p {
```

```
display: flex;
justify-content: space-between;
margin: 10px 0;
font-size: 17px;
}
```

```
.total {
border-top: 2px solid #333;
padding-top: 12px;
font-size: 21px !important;
font-weight: bold;
color: #2a9d8f;
}
```

```
.checkout {
width: 100%;
margin-top: 15px;
background: #2a9d8f;
}
```

```
@media (max-width: 700px) {
body {
padding: 15px;
}
```

```
.product-form {
grid-template-columns: 1fr;
}
```

```
.container {  
    padding: 20px;  
}  
  
table {  
    font-size: 13px;  
}  
  
th, td {  
    padding: 8px;  
}  
}  
  
</style>  
</head>  
  
<body>  
  
<div class="container">  
  
<h1>🛒 Shopping Cart & Billing</h1>  
  
<div class="product-form">  
    <input type="text" id="productName" placeholder="Product Name">  
    <input type="number" id="price" placeholder="Price">  
    <input type="number" id="quantity" placeholder="Quantity" min="1" value="1">  
    <button onclick="addProduct()">Add Product</button>  
</div>
```

```
<table>
  <thead>
    <tr>
      <th>Product</th>
      <th>Price</th>
      <th>Quantity</th>
      <th>Subtotal</th>
      <th>Action</th>
    </tr>
  </thead>

  <tbody id="cartTable"></tbody>
</table>
```

```
<div class="billing">
  <p>
    <span>Subtotal:</span>
    <span>₹<span id="subtotal">0.00</span></span>
  </p>
```

```
<p>
  <span>Discount (10%):</span>
  <span>₹<span id="discount">0.00</span></span>
</p>
```

```
<p>
  <span>Tax (18%):</span>
```

```
    <span>₹<span id="tax">0.00</span></span>
  </p>
```

```
<p class="total">
  <span>Total:</span>
  <span>₹<span id="total">0.00</span></span>
</p>
```

```
<button class="checkout" onclick="checkout()">Checkout</button>
</div>
```

```
</div>
```

```
<script>
```

```
  let cart = [];
```

```
  function addProduct() {
```

```
    const name = document.getElementById("productName").value.trim();
```

```
    const price = Number(document.getElementById("price").value);
```

```
    const quantity = Number(document.getElementById("quantity").value);
```

```
    if (name === "" || price <= 0 || quantity <= 0) {
```

```
      alert("Please enter valid product details.");
```

```
      return;
```

```
    }
```

```
    const product = {
```

```
      name: name,
```



```
    price: price,  
    quantity: quantity  
  };  
  
  cart.push(product);  
  
  document.getElementById("productName").value = "";  
  document.getElementById("price").value = "";  
  document.getElementById("quantity").value = 1;  
  
  displayCart();  
  calculateBill();  
}
```

```
function displayCart() {  
  const table = document.getElementById("cartTable");  
  table.innerHTML = "";  
  
  cart.forEach(function(product, index) {  
    const subtotal = product.price * product.quantity;  
  
    const row = `  
      <tr>  
        <td>${product.name}</td>  
        <td>₹${product.price.toFixed(2)}</td>  
        <td>  
          <input  
            class="quantity"
```

```

        type="number"
        min="1"
        value="${product.quantity}"
        onchange="updateQuantity(${index}, this.value)"
    >
</td>
<td>₹${subtotal.toFixed(2)}</td>
<td>
    <button class="remove" onclick="removeProduct(${index})">
        Remove
    </button>
</td>
</tr>
`;

table.innerHTML += row;
});
}

function updateQuantity(index, quantity) {
    quantity = Number(quantity);

    if (quantity < 1) {
        quantity = 1;
    }

    cart[index].quantity = quantity;

```

```
    displayCart();  
    calculateBill();  
}
```

```
function removeProduct(index) {  
    cart.splice(index, 1);  
  
    displayCart();  
    calculateBill();  
}
```

```
function calculateBill() {  
    let subtotal = 0;  
  
    cart.forEach(function(product) {  
        subtotal += product.price * product.quantity;  
    });
```

```
    let discount = subtotal * 0.10;  
    let taxableAmount = subtotal - discount;  
    let tax = taxableAmount * 0.18;  
    let total = taxableAmount + tax;
```

```
    document.getElementById("subtotal").innerText =  
        subtotal.toFixed(2);
```

```
    document.getElementById("discount").innerText =  
        discount.toFixed(2);
```

```
document.getElementById("tax").innerText =  
    tax.toFixed(2);
```

```
document.getElementById("total").innerText =  
    total.toFixed(2);  
}
```

```
function checkout() {  
    if (cart.length === 0) {  
        alert("Your cart is empty.");  
        return;  
    }
```

```
    const total = document.getElementById("total").innerText;
```

```
    alert("Order placed successfully!\nTotal Amount: ₹" + total);
```

```
    cart = [];  
    displayCart();  
    calculateBill();  
}
```

```
displayCart();  
calculateBill();
```

```
</script>
```

```
</body>
```

